

Emmanuel Tega Agbragu

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SUMMARY

Security Engineer with years of experience across offensive and defensive security, specializing in network security, threat analysis, and automation. Proven track record in analyzing emerging security threats, performing meticulous data analysis, and developing prevention logic for enterprise security infrastructure. Expert in networking protocols and scripting to enhance cloud security posture and automate internal tools.

EXPERIENCE

Security Researcher

Jan 2025 – Apr 2026

The Upanzi Network, Carnegie Mellon University | Kigali, Rwanda

- Ran web and API penetration tests across **5** research platforms, utilizing Wireshark for deep packet inspection to uncover vulnerabilities and deliver reproducible proof-of-concept evidence.
- Chained a mass-assignment flaw with a BOLA/IDOR vulnerability rated **Critical (CVSS 9.1)** to demonstrate full unauthorized modifications on an e-commerce API, quantifying worst-case impact for remediation prioritization.
- Identified an indirect prompt-injection vulnerability (OWASP **LLM01**) in an LLM-powered dataset feature, researching and proving model-behavior manipulation via crafted PDF uploads to surface AI-specific risks.
- Authored OWASP- and NIST-aligned VAPT reports for each engagement and drove remediation through to verified closure, retesting fixes with development teams to ensure robust security.
- Built custom Python automation scripts and extended APKit to scale endpoint enumeration and vulnerability detection across large asset inventories, improving research efficiency by **30%**.

Graduate Teaching Assistant

Sep 2024 – Jan 2025

Carnegie Mellon University Africa | Kigali, Rwanda

- Delivered lab sessions in Ethical Hacking and Cybersecurity Operations, reinforcing student competency in offensive and defensive security, networking architecture, and protocol analysis.

Cybersecurity Engineer

Apr 2021 – Jun 2023

Lumenave International Limited | Lagos, Nigeria

- Designed and deployed security infrastructure including IDS/IPS and Firewall solutions for banking clients, reducing malware infections by **40%** through rigorous system hardening and signature development.
- Implemented CyberArk PAM solutions with role-based access controls and least-privilege policies to meet SOC 2 and ISO 27001 compliance requirements for regulated network environments.
- Configured Symantec DLP policies defining detection rules and access controls to prevent unauthorized exfiltration of sensitive financial data across SSL and SMB protocols.
- Led a team of 5 engineers in the management of security projects, ensuring on-time delivery within strict budget and technical constraints.
- Oversaw security operations for all deployed solutions, providing technical support and PowerShell automation that maintained **100%** operational efficiency.

SELECTED PROJECTS

Securing the Digital State — Research Report

2025

Python, Automation | Upanzi Network

- Contributed to a pan-African study assessing the security posture of **21,782** government sub-domains across 54 countries.
- Led HTTP-methods enumeration and protocol analysis, identifying dangerous exposed methods and network security gaps.
- Built custom Python automation scripts powering mass asset discovery and scanning across government assets.

Privileged Access Management Deployment

2021 – 2023

CyberArk, Bash, Oracle, PowerShell | First City Monument Bank

- Deployed a complete CyberArk stack managing privileged account access across network perimeters, with secure authentication and least-privilege role-based access controls aligned to regulatory requirements.

TECHNICAL SKILLS

Security Tools	Burp Suite, OWASP ZAP, Nmap, Nessus, OpenVAS, SQLMap, IDS/IPS
Web & Networking	HTTP, TCP/IP, REST APIs, OAuth2, SAML, OIDC, LDAP, Kerberos, DNS
Programming	Python, Bash, Go, Java, SQL, PHP
Cloud & Infrastructure	AWS, Terraform, CloudFormation, CI/CD, Docker
Frameworks	OWASP Top 10, MITRE ATT&CK, NIST, ISO 27001, SOC 2, PCI-DSS

EDUCATION

MSc Information Technology

Aug 2023 – May 2025

Carnegie Mellon University | GPA: 3.80/4.0 | Mastercard Foundation Partial Scholarship

BSc Pure and Applied Physics, First Class Honours

Mar 2014 – May 2018

University of Benin | Nigeria | GPA: 4.54/5.0 | Top 1% of graduating class

CERTIFICATIONS AND ACHIEVEMENTS

CEH Master • eJPT • AWS Certified Cloud Practitioner • OSCP (In Progress) • Top 10 Finalist, Greenfist CTF 2023 • Mastercard Foundation Scholarship 2023 • First Class Honours, Top 1%, 2018